

OS LAB 10

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Task 1 (Code)

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <ctype.h>
#include <sys/wait.h>

int main()
{
    int fd[2];
    pid_t pid;
    char msg[100];

    pipe(fd);
    pid = fork();

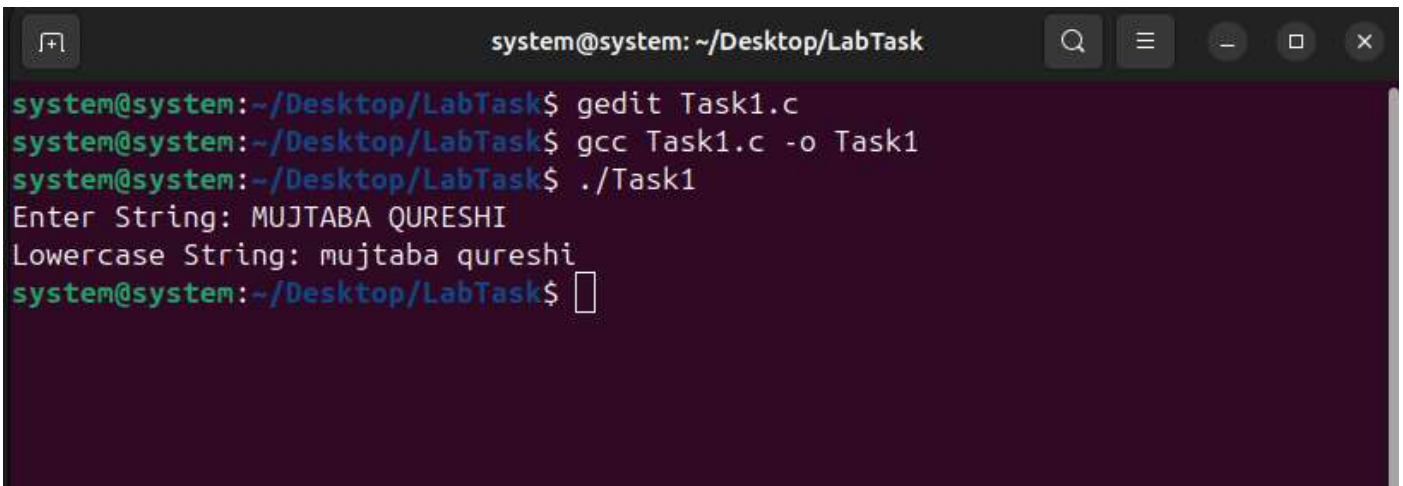
    if(pid > 0)
    {
        close(fd[0]);
        printf("Enter String: ");
        fgets(msg, sizeof(msg), stdin);
        write(fd[1], msg, strlen(msg) + 1);
        close(fd[1]);
        wait(NULL);
    }
    else
    {
        close(fd[1]);
        read(fd[0], msg, sizeof(msg));

        for(int i = 0; msg[i] != '\0'; i++)
            msg[i] = tolower(msg[i]);

        printf("Lowercase String: %s", msg);
        close(fd[0]);
    }

    return 0;
}
```

Task 1 (Screenshot)



```
system@system: ~/Desktop/LabTask
system@system:~/Desktop/LabTask$ gedit Task1.c
system@system:~/Desktop/LabTask$ gcc Task1.c -o Task1
system@system:~/Desktop/LabTask$ ./Task1
Enter String: MUJTABA QURESHI
Lowercase String: mujtaba qureshi
system@system:~/Desktop/LabTask$
```

Task 2 (Code)

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int shmid;
    int *balance;

    shmid = shmget(IPC_PRIVATE, sizeof(int), IPC_CREAT | 0666);
    balance = (int*)shmat(shmid, NULL, 0);

    *balance = 5000;

    if(fork() == 0)
    {
        *balance += 2000;
        printf("Updated Balance: %d\n", *balance);
    }
    else
    {
        wait(NULL);
        printf("Read Balance: %d\n", *balance);
    }

    shmdt(balance);
    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

Task 2 (Screenshot)

```
system@system:~/Desktop/LabTask$ gedit Task2.c
system@system:~/Desktop/LabTask$ gcc Task2.c -o Task2
system@system:~/Desktop/LabTask$ ./Task2
Updated Balance: 10000
Read Balance: 10000
system@system:~/Desktop/LabTask$
```

Task 3 (Code)

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

char data[100];

void *download(void *arg)
{
    printf("Downloading Data...\n");
    sleep(2);
    sprintf(data, "Server File Downloaded");
    printf("Download Complete!\n");
    pthread_exit(NULL);
}

void *process(void *arg)
{
    sleep(3);
    printf("Processing Data: %s\n", data);
    pthread_exit(NULL);
}

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, download, NULL);
    pthread_create(&t2, NULL, process, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    return 0;
}
```

Task 3 (Screenshot)

```
system@system:~/Desktop/LabTask$ gedit Task3.c
system@system:~/Desktop/LabTask$ gcc Task3.c -o Task3 -lpthread
system@system:~/Desktop/LabTask$ ./Task3
Downloading Data...
Download Complete!
Processing Data: Server File Downloaded
system@system:~/Desktop/LabTask$
```

Task 4 (Code)

```
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

void handle_sigint(int sig)
{
    printf("\nCtrl+C Pressed! Process will continue.\n");
}

void handle_sigchld(int sig)
{
    printf("Child Process Terminated.\n");
}

int main()
{
    signal(SIGINT, handle_sigint);
    signal(SIGCHLD, handle_sigchld);

    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Running...\n");
        sleep(5);
    }
    else
    {
        sleep(2);
        kill(pid, SIGKILL);

        while(1)
        {
            printf("Parent Running... Press Ctrl+C\n");
            sleep(2);
        }
    }

    return 0;
}
```

Task 4 (Screenshot)

```
system@system:~/Desktop/LabTask$ gedit Task4.c
system@system:~/Desktop/LabTask$ gcc Task4.c -o Task4
system@system:~/Desktop/LabTask$ ./Task4
Child Running...
Parent Running... Press Ctrl+C
Child Process Terminated.
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C

Ctrl+C Pressed! Process will continue.
Parent Running... Press Ctrl+C
^CParent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
^C
Ctrl+C Pressed! Process will continue.
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
Parent Running... Press Ctrl+C
```

Task 5 (Code)

```
#include <pthread.h>
#include <stdio.h>
#include <unistd.h>

#define USERS 5

pthread_mutex_t mutex;

void *printer(void *arg)
{
    int id = *((int*)arg);

    pthread_mutex_lock(&mutex);

    printf("User %d is Printing...\n", id);
    sleep(2);
    printf("User %d Finished Printing.\n", id);

    pthread_mutex_unlock(&mutex);

    pthread_exit(NULL);
}

int main()
{
    pthread_t threads[USERS];
```

```
int ids[USERS];

pthread_mutex_init(&mutex, NULL);

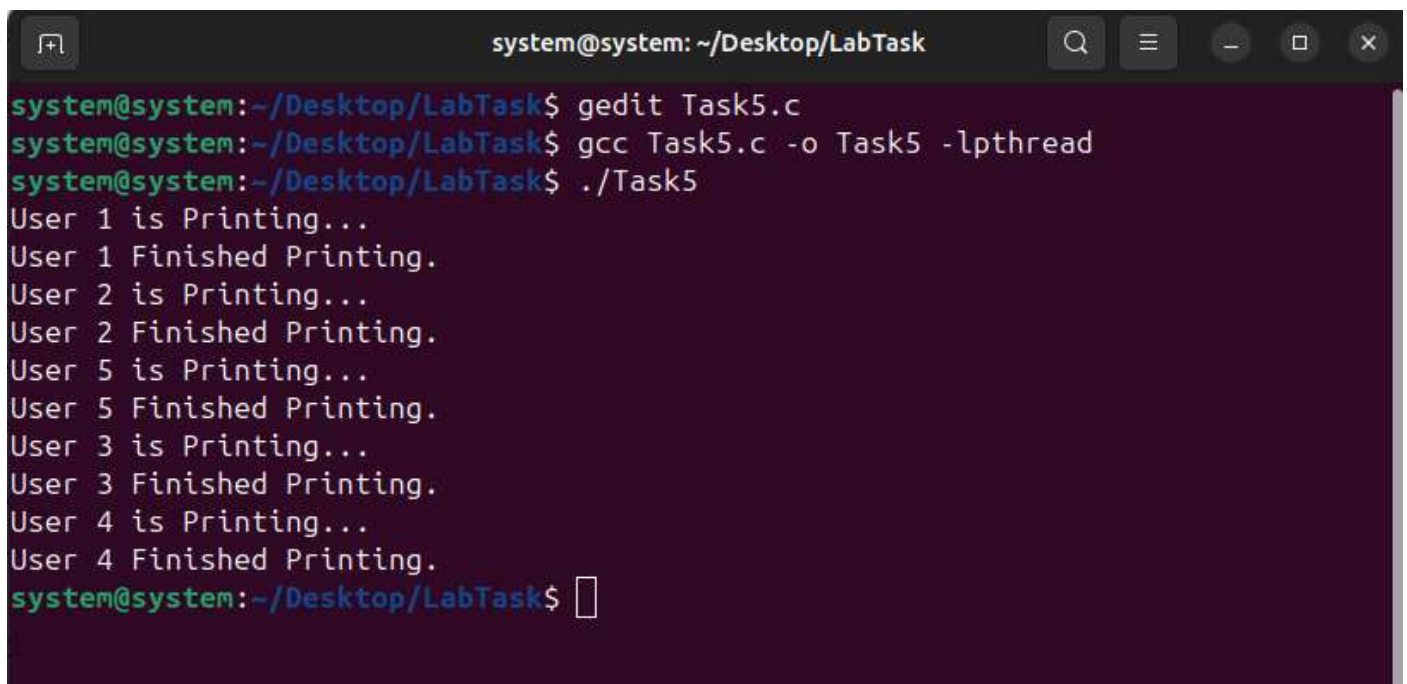
for(int i = 0; i < USERS; i++)
{
    ids[i] = i + 1;
    pthread_create(&threads[i], NULL, printer, &ids[i]);
}

for(int i = 0; i < USERS; i++)
    pthread_join(threads[i], NULL);

pthread_mutex_destroy(&mutex);

return 0;
}
```

Task 5 (Screenshot)



The screenshot shows a terminal window titled "system@system: ~/Desktop/LabTask". The user has executed the following commands:

```
system@system:~/Desktop/LabTask$ gedit Task5.c
system@system:~/Desktop/LabTask$ gcc Task5.c -o Task5 -lpthread
system@system:~/Desktop/LabTask$ ./Task5
```

The output of the program is as follows:

```
User 1 is Printing...
User 1 Finished Printing.
User 2 is Printing...
User 2 Finished Printing.
User 5 is Printing...
User 5 Finished Printing.
User 3 is Printing...
User 3 Finished Printing.
User 4 is Printing...
User 4 Finished Printing.
system@system:~/Desktop/LabTask$
```

The output shows that the program successfully executed and printed the messages for all five users, with the order of completion being 1, 2, 5, 3, and 4.

